Canonical involutions

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Let E be a set and $(E \times E, \pi_1, \pi_2)$ be the product of E with E.

Show that there is a unique mapping $\tau : E \times E \longrightarrow E \times E$ such that $\pi_1 \circ \tau = \pi_2$ and $\pi_2 \circ \tau = \pi_1$.

Compute τ^2 and conclude that τ is a bijection.

Let E be a set.

Let $(E \amalg E, i_1, i_2)$ be the coproduct of E with E.

Show that there is a unique mapping $\sigma : E \times E \longrightarrow E \times E$ such that $\sigma \circ i_1 = i_2$ and $\sigma \circ i_2 = i_1$.

Compute σ^2 and conclude that σ is a bijection.